Module 2:Transport Layer and Network Layer

Module 2:Transport Layer and Network Layer

Transport Layer

- Services and Protocols
- UDP and TCP

Hands-on

- Sockets Introduction
- Elementary TCP Sockets
- TCP Client/Server Example
- I/O Multiplexing
  - select()
  - poll()
- Elementary UDP Sockets
- Advanced I/O Functions

Network Layer

- Introduction
- Network-layer Protocols
- Unicast Routing
- Multicast Routing
- Multicasting Basics
- Intra-domain Routing
- Inter-domain Routing
- Next Generation IP (IPv6)
- Quality of Service (QoS)

Hands-on

- Linux Routing Table
- Routing Caches
- Routing Cache Implementation
- Adding Routes using `ip`

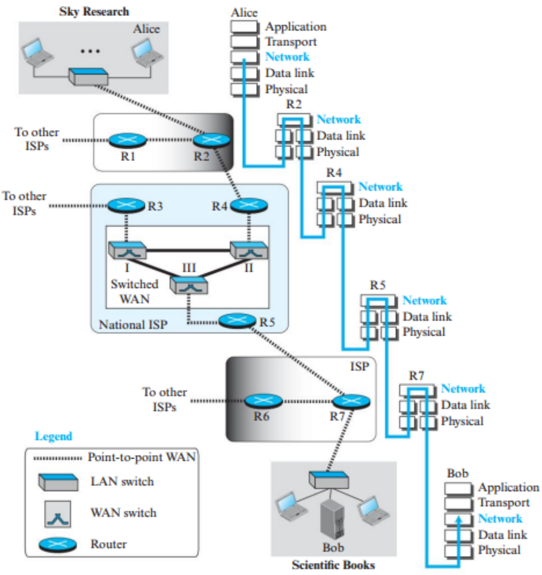
4.1 Introduction to the Network Layer

- Communication between Alice and Bob at the **Network Layer**.
- The Internet is an **internetwork** consisting of multiple **LANs and WANs**.
- These networks are interconnected through **routers (or switches)**.
- The Network Layer provides communication across multiple interconnected networks.

Key Idea

The Network Layer enables end-to-end packet delivery across an internetwork by forwarding packets through intermediate routers.

Figure 4.1 *Communication at the network layer*



Role of the Network Layer

Network Layer operates at:

- **Source Host (Alice)**
  - Receives packets from the Transport Layer.
  - Encapsulates packets into IP datagrams.
  - Passes datagrams to the Data-Link Layer.
- **Routers**
  - Forward datagrams between interconnected networks.
  - Normally operate using the Physical, Data-Link and Network layers.
  - May use Transport/Application layers for network control.
- **Destination Host (Bob)**
  - Decapsulates the received datagram.
  - Extracts the packet.
  - Delivers it to the appropriate Transport Layer.

4.1.1 Network Layer Services

Major Services of the Network Layer

1. Packetizing
2. Routing
3. Forwarding
4. Error Control
5. Flow Control
6. Congestion Control
7. Quality of Service (QoS)
8. Security

Goal

Provide end-to-end packet delivery across interconnected networks efficiently and securely.

Packetizing, Routing and Forwarding

Packetizing

- Encapsulates transport-layer payload into a network-layer packet.
- Source adds IP header.
- Destination decapsulates and delivers payload.
- Supports fragmentation and reassembly when required.

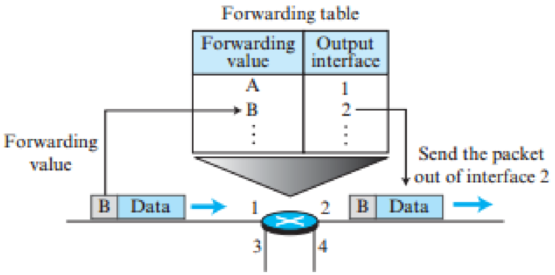
Routing

- Determines the best path from source to destination.
- Uses routing protocols to build routing tables.

Forwarding

- Router forwards packets using the forwarding (routing) table.
- Decision is based on destination address or label.

Figure 4.2 *Forwarding process*



Error Control, Flow Control and Congestion Control

Error Control

- Internet Network Layer provides only **header checksum**.
- Payload errors are not checked.
- ICMP reports packet delivery errors.

Flow Control

- Not provided by the Network Layer.
- Managed by upper-layer protocols (e.g., TCP).

Congestion Control

- Prevents excessive traffic from overwhelming routers.
- Packet loss due to congestion may trigger retransmissions.
- Internet mainly handles congestion at upper layers.

Quality of Service (QoS)

- Supports multimedia applications (audio/video).
- Focuses on delay, bandwidth and reliability.
- Mostly implemented by upper-layer protocols.

Security

- Original IP provided no security.
- Modern Internet uses **IPSec** to provide:
  - Authentication
  - Confidentiality
  - Data integrity

4.1.2 Packet Switching

- Network-layer communication uses **Packet Switching**.
- Messages are divided into smaller **packets**.
- Routers forward packets from input ports to output ports.
- Destination reassembles all packets before delivering the message.

Packet Switching Approaches

1. Datagram (Connectionless)
2. Virtual Circuit (Connection-Oriented)

Datagram Approach (Connectionless)

- Each packet is treated independently.
- No connection is established before transmission.
- Packets belonging to the same message may follow different paths.
- Routers make forwarding decisions using the **destination address**.
- Source address is mainly used for error reporting.

Figure 4.3 *A connectionless packet-switched network*

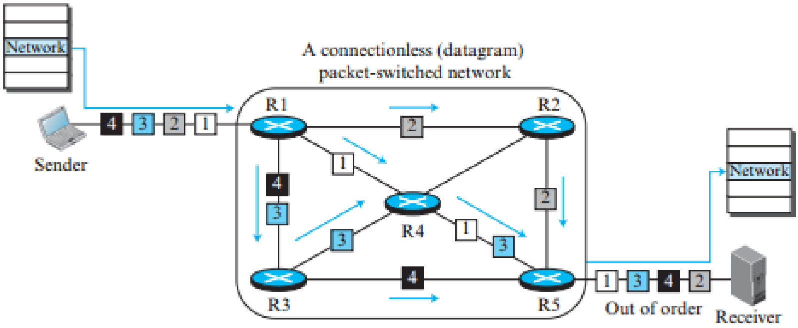
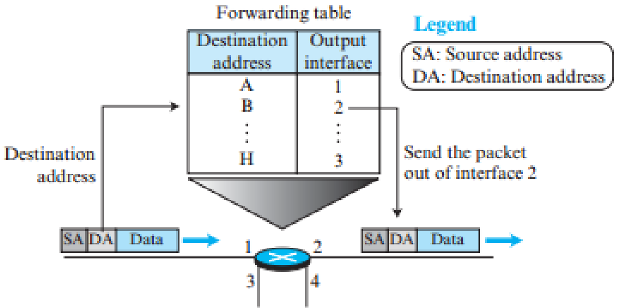


Figure 4.4 *Forwarding process in a router when used in a connectionless network*



Virtual Circuit Approach (Connection-Oriented)

- A virtual path is established before data transfer.
- All packets follow the same route.
- Each packet carries a **Virtual Circuit Identifier (Label)**.
- Routers forward packets based on the label instead of only the destination address.
- Communication occurs in three phases:
  1. Setup
  2. Data Transfer
  3. Teardown

Figure 4.5 *A virtual-circuit packet-switched network*

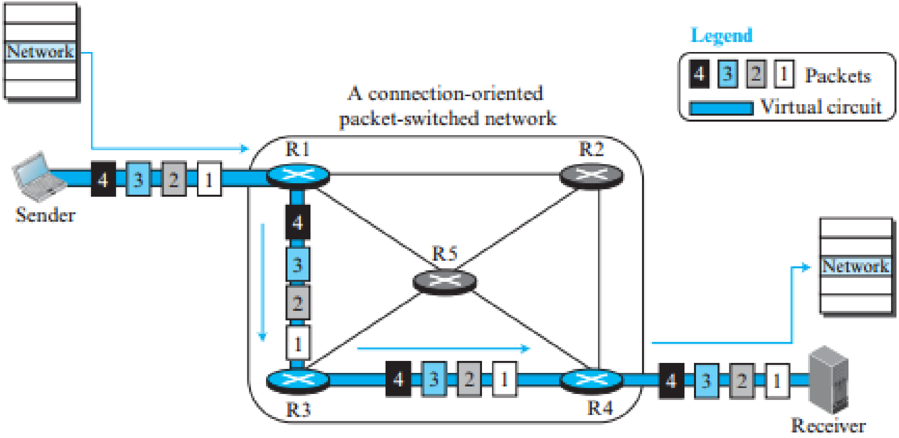
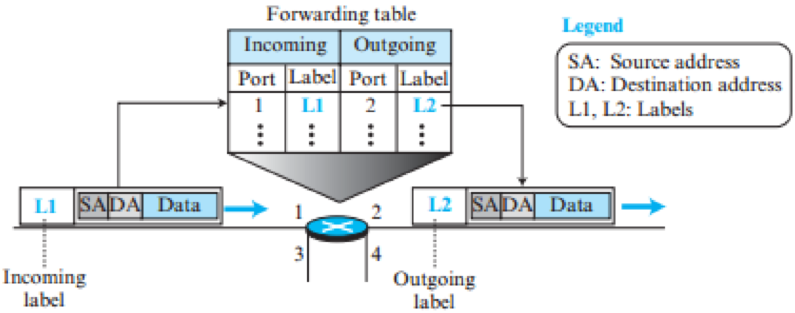


Figure 4.6 *Forwarding process in a router when used in a virtual-circuit network*



Virtual Circuit Setup Phase

Setup Process

1. Source sends a **Request Packet**.
2. Each router:
   - Creates a forwarding-table entry.
   - Assigns incoming port, label and outgoing port.
3. Destination assigns the final incoming label.
4. Acknowledgment packet travels back.
5. Routers complete forwarding-table entries.

Figure 4.7 *Sending request packet in a virtual-circuit network*

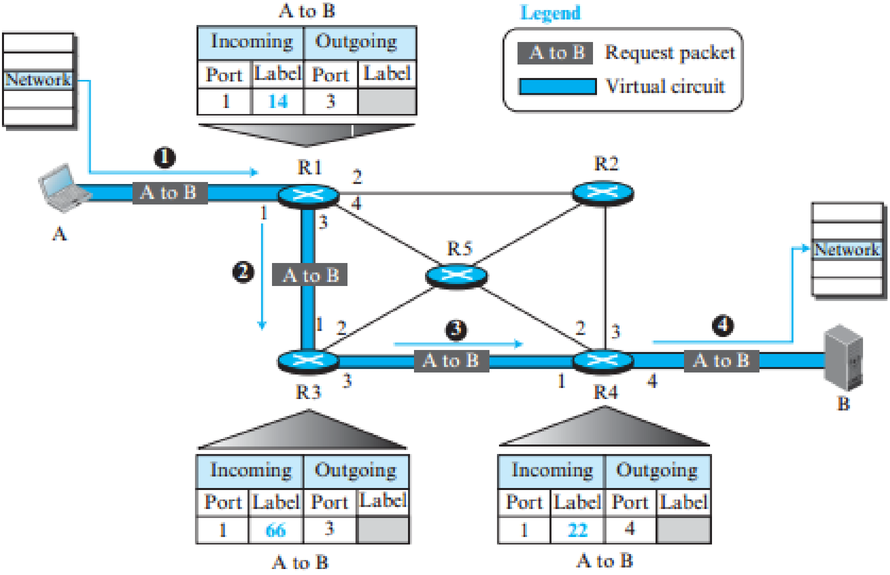
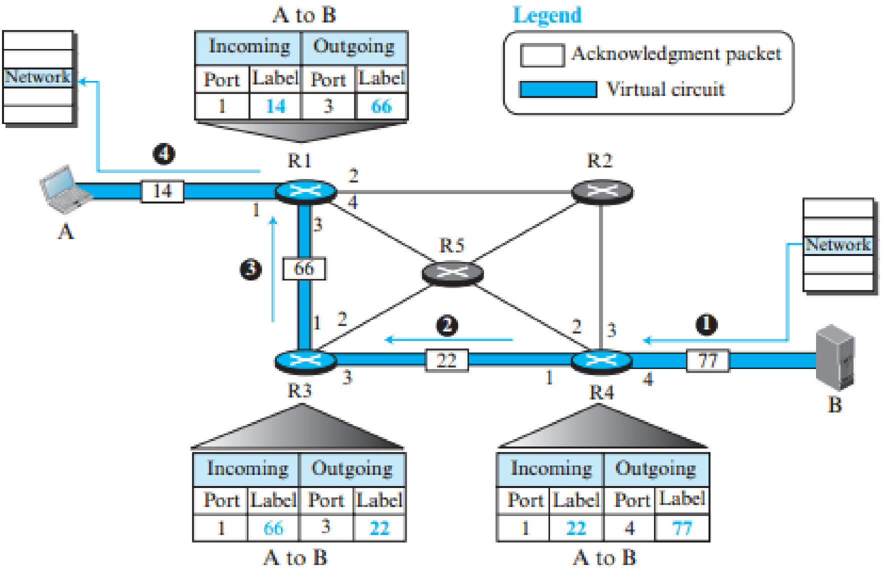


Figure 4.8 *Sending acknowledgments in a virtual-circuit network*



A to B (R1)

Incoming		Outgoing	
Port	Label	Port	Label
1	14	3	66

A to B (R3)

Incoming		Outgoing	
Port	Label	Port	Label
1	66	3	22

A to B (R4)

Incoming		Outgoing	
Port	Label	Port	Label
1	22	4	77

Data Transfer and Teardown

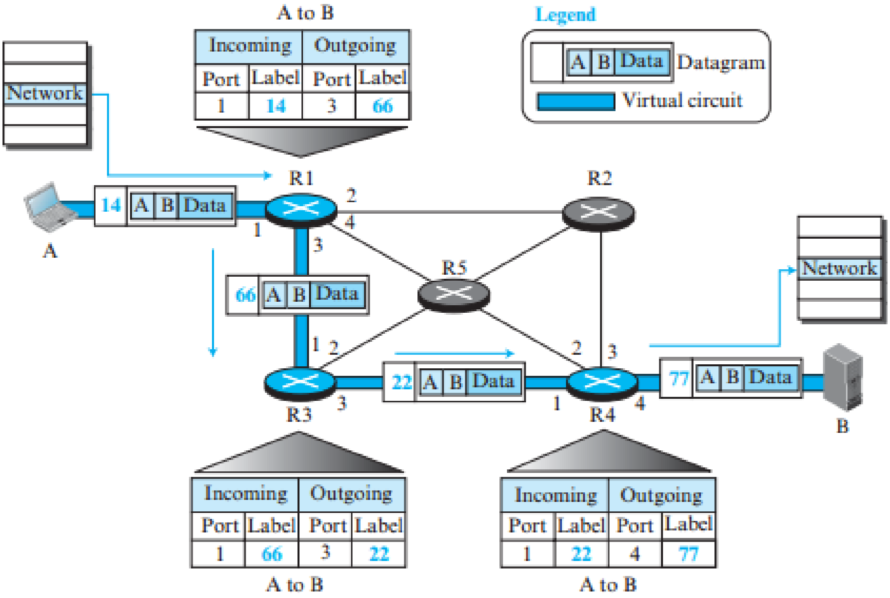
Data Transfer

- Source sends packets using the assigned label.
- Each router replaces the incoming label with a new outgoing label.
- All packets follow the same virtual path.
- Packets arrive in order at the destination.

Teardown

- Source sends a teardown packet.
- Destination sends confirmation.
- Routers delete forwarding-table entries.

Figure 4.9 *Flow of one packet in an established virtual circuit*



Datagram vs Virtual Circuit

Datagram	**Virtual Circuit**
Connectionless	Connection-oriented
No setup required	Setup required
Packets routed independently	All packets follow same path
Uses destination address	Uses virtual circuit label
Packets may arrive out of order	Packets arrive in order
Simple and flexible	Better flow management

4.1.3 Network Layer Performance

Performance Metrics

The performance of a packet-switched network is evaluated using three major metrics:

1. Delay
2. Throughput
3. Packet Loss

Objective

A good network minimizes delay and packet loss while maximizing throughput.

Delay in Packet-Switched Networks

A packet experiences four types of delay:

1. **Transmission Delay**

$$Delay_{tr} = \frac{\text{Packet Length}}{\text{Transmission Rate}}$$

2. **Propagation Delay**

$$Delay_{pg} = \frac{\text{Distance}}{\text{Propagation Speed}}$$

3. **Processing Delay**
   - Header processing
   - Error checking
   - Forwarding decision
4. **Queuing Delay**
   - Waiting time in router input/output queues

Total Delay

End-to-End Delay

$$\boxed{Total\ Delay = (n+1)(Delay_{tr} + Delay_{pg} + Delay_{pr}) + n(Delay_{qu})}$$

where

- n = Number of routers
- $(n+1)$ = Number of links

Example

- Fast Ethernet (100 Mbps)
- Packet Size = 10,000 bits

$$Delay_{tr} = \frac{10000}{100 \times 10^6} = 100\ \mu s$$

Throughput

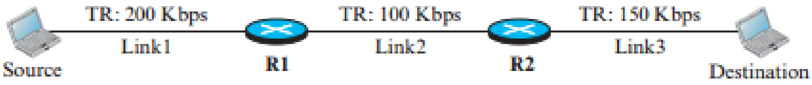
Throughput

- Number of bits transmitted per second.
- Determined by the **slowest link (bottleneck)**.

$$\boxed{Throughput = \min\{TR_1, TR_2, \ldots, TR_n\}}$$

- Internet throughput is usually limited by the access links.
- Shared links divide available bandwidth among multiple flows.

Figure 4.10 *Throughput in a path with three links in a series*



a. A path through three links

TR: Transmission rate

b. Simulation using pipes

Figure 4.11 *A path through the Internet backbone*

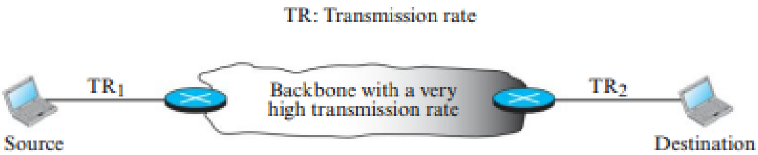
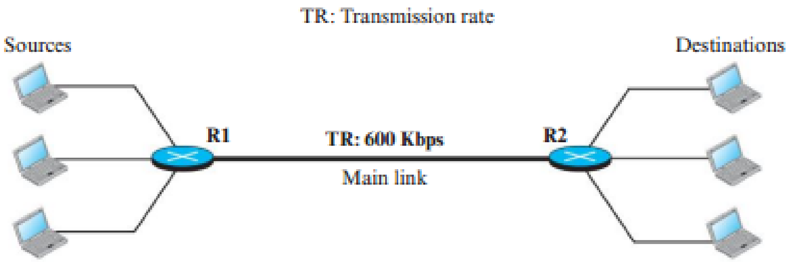


Figure 4.12 *Effect of throughput in shared links*



Packet Loss

Packet Loss

- Occurs when router buffers become full.
- Excess packets are discarded.
- Lost packets must be retransmitted.
- Frequent retransmissions increase congestion.
- Severe congestion may lead to network collapse.

Summary

- Low Delay $\Rightarrow$ Faster communication
- High Throughput $\Rightarrow$ Better bandwidth utilization
- Low Packet Loss $\Rightarrow$ Reliable communication

4.1.4 Network-Layer Congestion

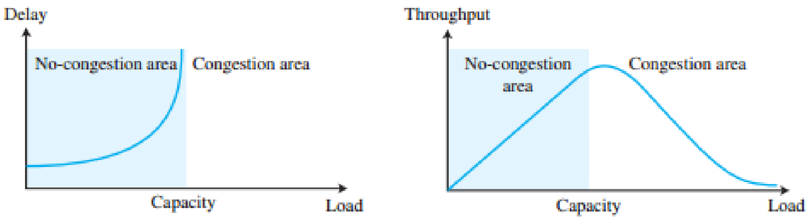
Network Congestion

Congestion occurs when the offered load exceeds the network capacity, causing:

- Increased delay
- Packet loss
- Reduced throughput

- Delay rises sharply after network capacity.
- Throughput increases initially, then decreases due to packet drops and retransmissions.

Figure 4.13 *Packet delay and throughput as functions of load*



a. Delay as a function of load

b. Throughput as a function of load

Congestion Control Techniques

Types of Congestion Control

1. **Open-Loop Control (Prevention)**
   - Prevent congestion before it occurs.
2. **Closed-Loop Control (Removal)**
   - Detect and reduce congestion after it occurs.

Open-Loop Congestion Control

Policies to Prevent Congestion

- **Retransmission Policy**
  - Efficient retransmission avoids unnecessary traffic.
- **Window Policy**
  - Selective Repeat is preferred over Go-Back-N.
- **Acknowledgment Policy**
  - Delayed or cumulative ACKs reduce network load.
- **Discarding Policy**
  - Drop less important packets during congestion.
- **Admission Policy**
  - Admit new virtual circuits only if resources are available.

Closed-Loop Congestion Control

Techniques to Remove Congestion

- **Backpressure**
  - Congested router asks upstream routers to slow down.
  - Used only in virtual-circuit networks.
- **Choke Packet**
  - Router sends warning directly to the source.
- **Implicit Signaling**
  - Sender infers congestion from delayed ACKs or timeout.
- **Explicit Signaling**
  - Congestion information carried within data packets.

Figure 4.14 *Backpressure method for alleviating congestion*

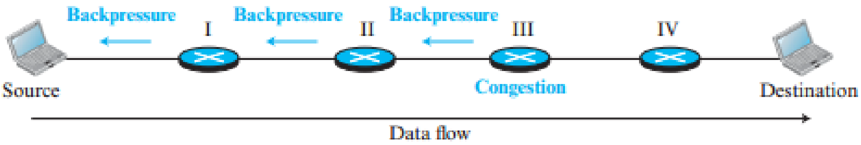
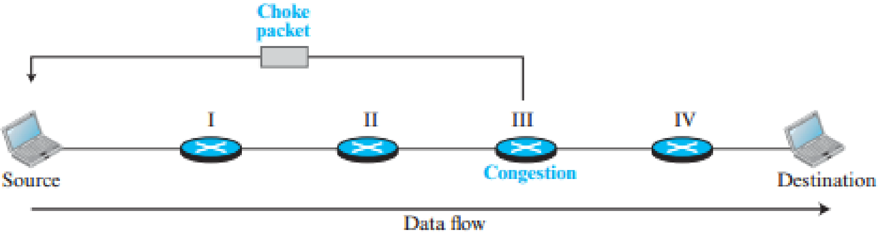


Figure 4.15 *Choke packet*



Congestion Control Summary

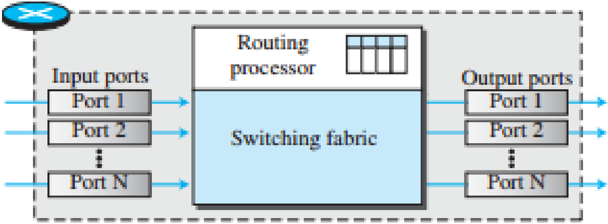
Method	**Purpose**
Retransmission Policy	Reduce unnecessary retransmissions
Window Policy	Prefer Selective Repeat
Acknowledgment Policy	Reduce ACK traffic
Discarding Policy	Drop low-priority packets
Admission Policy	Prevent overload
Backpressure	Notify upstream routers
Choke Packet	Warn source directly
Implicit Signaling	Detect congestion via timeout/ACK delay
Explicit Signaling	Carry congestion information in packets

4.1.5 Structure of a Router

Main Components of a Router

1. Input Ports
2. Output Ports
3. Routing Processor
4. Switching Fabric

Figure 4.16 *Router components*



Router Components

Input Port

- Performs Physical and Data-Link layer functions.
- Decapsulates frames and checks errors.
- Stores incoming packets in input queues.

Output Port

- Queues outgoing packets.
- Encapsulates packets into frames.
- Performs Physical layer transmission.

Routing Processor

- Performs forwarding table lookup.
- Determines next hop and output port.
- Modern routers often perform lookup at input ports.

Figure 4.17 *Input port*

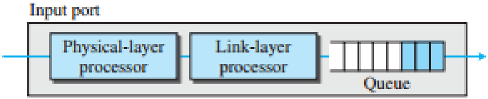


Figure 4.18 *Output port*

Switching Fabrics

Purpose: Move packets efficiently from input queues to output queues.

- **Crossbar Switch**
  - Grid of electronic switches.
  - Directly connects n inputs to n outputs.
- **Banyan Switch**
  - Multi-stage switching network.
  - Routes packets using destination binary bits.
  - Requires $\log_2(n)$ stages.

Figure 4.19 *Crossbar switch*

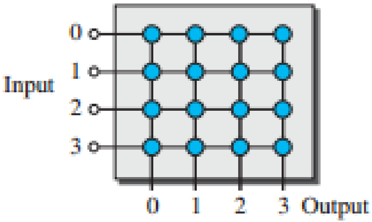


Figure 4.20 *Banyan switch*

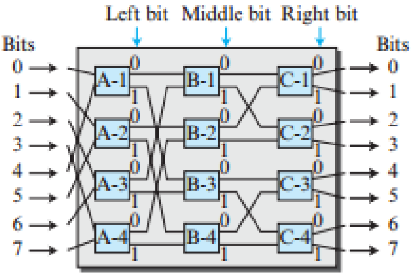
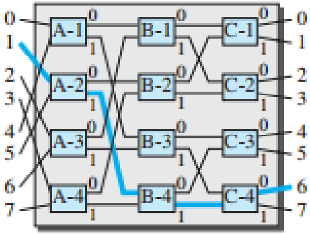
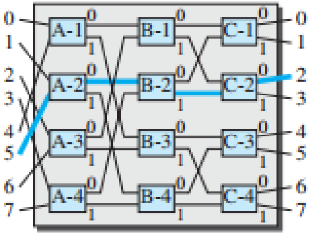


Figure 4.21 *Examples of routing in a banyan switch*



a. Input 1 sending to output 6 (110)



b. Input 5 sending to output 2 (010)

Batcher-Banyan Switch

Need

Banyan switches may suffer from **internal collisions**.

- **Batcher Switch**
  - Sorts packets by destination before switching.
- **Trap Module**
  - Prevents multiple packets with the same destination from entering simultaneously.
- **Batcher-Banyan Switch**
  - Combines sorting and Banyan switching.
  - Improves efficiency and avoids collisions.

Figure 4.22 *Batcher-banyan switch*